

Renewable Hydride Donors for the Catalytic Reduction of CO₂: A Thermodynamic and Kinetic Study

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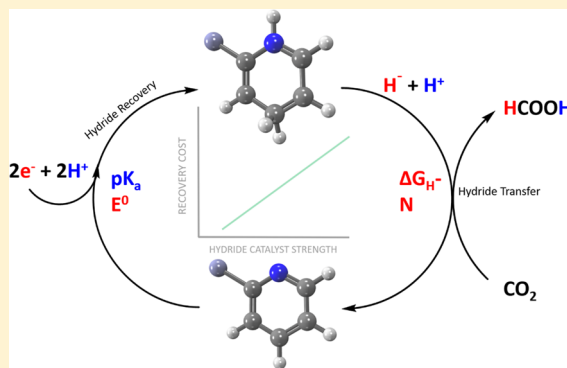
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S Supporting Information

ABSTRACT: Increasing atmospheric CO₂ concentration and dwindling fossil fuel supply necessitate the search for efficient methods for CO₂ conversion to fuels. Assorted studies have shown pyridine and its derivatives capable of (photo)electrochemically reducing CO₂ to methanol, and some mechanistic interpretations have been proposed. Here, we analyze the thermodynamic and kinetic aspects of the efficacy of pyridines as hydride-donating catalytic reagents that transfer hydrides via their dihydropyridinic form. We investigate both the effects of functionalizing pyridinic derivatives with electron-donating and electron-withdrawing groups on hydride-transfer catalyst strength, assessed via their hydricity (thermodynamic ability) and nucleophilicity (kinetic ability), and catalyst recyclability, assessed via their reduction potential. We find that pyridines substituted with electron-donating groups have stronger hydride-donating ability (having lower hydricity and larger nucleophilicity values), but are less efficiently recycled (having more negative reduction potentials). In contrast, pyridines substituted with electron-withdrawing groups are more efficiently recycled, but are weaker hydride donors. Functional group modification favorably tunes hydride strength or efficiency, but not both. We attribute this problematic coupling between the strength and recyclability of pyridinic hydrides to their aromatic nature and suggest several avenues for overcoming this difficulty.

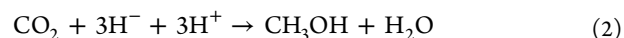
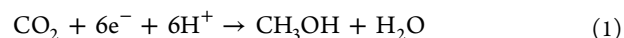


1. INTRODUCTION

Concerns about the impact of atmospheric carbon dioxide (CO₂) on the climate and ever rising global energy demands have spurred growing efforts toward efficient conversion of CO₂ into useful products, such as fuels (e.g., methanol).^{1–4} The solution to this problem via imitating natural CO₂ reduction has yet proven successful. Even the less ambitious goal of developing catalysts that efficiently transform CO₂ into valuable products is extremely challenging.^{3,5–9}

The conversion of CO₂ into valuable products by reducing CO₂ via a series of one-electron transfers (ETs) and proton transfers (PTs) produces open-shell (radical) high-energy intermediates at every odd electron reduction. This leads to slow kinetics and low selectivities except in cases where these radicals are stabilized.¹⁰ The large energy cost of producing these radical intermediates is demonstrated by the significantly negative reduction potential of –2.14 V vs saturated calomel electrode (SCE) for the one-electron reduction of CO₂ to CO₂^{•–}.^{11,12} Nature circumvents this difficulty by avoiding radical intermediates altogether in favor of closed-shell, stable

intermediates by performing reductions of two electron at a time as hydride (H[–]) transfers (HTs), which are effectively 2e[–]/H⁺ reductions.¹³ Consequently, the six ETs and six PTs that reduce CO₂ to methanol could in principle be accomplished as three HTs and three PTs, as represented by eqs 1 and 2.¹⁴



Dihydropyridines (DHPs) and their derivatives mimic nature's approach for reducing CO₂ through the NADP⁺/NADPH redox couple in the Calvin cycle of photosynthetic organisms.^{15–17} Our group recently reported the first detailed theoretical mechanism of converting CO₂ to methanol in aqueous solution catalyzed by 1,2-dihydropyridine (1,2-PyH₂) as a renewable organo-hydride.^{18,19} Each 1,2-PyH₂ transfers

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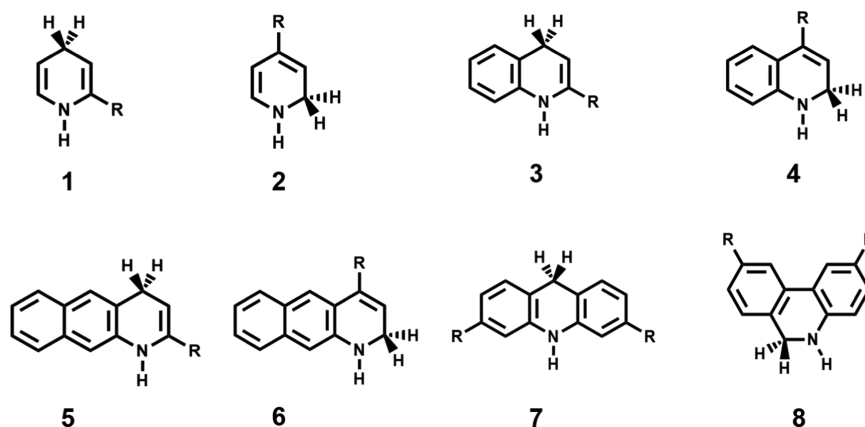


Figure 1. Dihydropyridine families examined in this work. Derivatives of 1,4-dihydropyridine (1), 1,2-dihydropyridine (2), 1,4-dihydroquinoline (3), 1,2-dihydroquinoline (4), 1,4-dihydrobenzoquinoline (5), 1,2-dihydrobenzoquinoline (6), dihydroacridine (7), and dihydrophenanthridine (8).

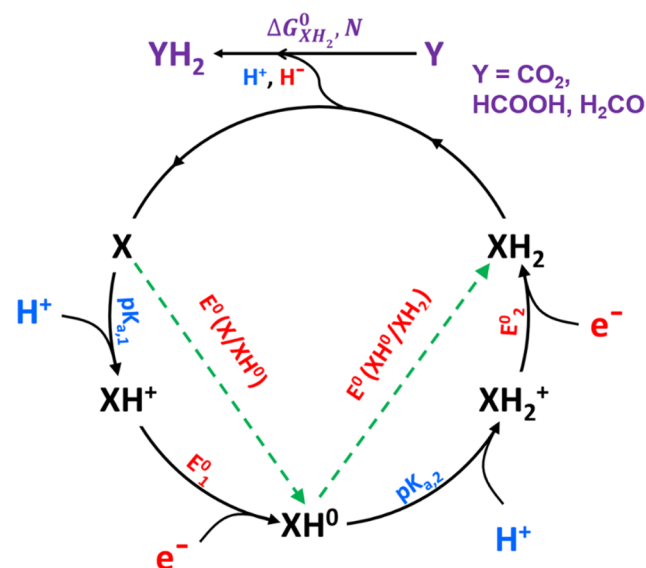
one hydridic and one protic hydrogen to CO_2 (sequentially) or its reduced products, formic acid and formaldehyde (concertedly). Thus, three molecules of 1,2-PyH₂ are required to convert CO_2 to methanol to satisfy eq 2. We also predicted that the PyH₂ DHP catalyst could be recycled via sequential PT–ET–PT–ET to pyridinic species in electrochemical, photochemical, and photoelectrochemical systems.¹⁹ Although our calculations predict that PyH₂ is a sufficiently strong hydride to reduce CO_2 to methanol homogeneously, this has not been observed experimentally, possibly indicating the existence of competitive side reactions. The mechanism proposed in refs 18 and 19 evidently does not produce methanol for all pyridine derivatives, such as pteridine, as observed by Tard et al.²⁰ and explained in ref 21. Different reaction conditions can allow other mechanistic pathways for the reduction of CO_2 to methanol by DHP derivatives;^{22,23} here we limit ourselves to the homogeneous catalysis by HTs.

In the present work, we focus on a theoretical procedure to aid in finding catalysts that function similarly to, but which outperform, 1,2-PyH₂ in facilitating both CO_2 reduction and efficient catalyst regeneration. To this end, we use quantum chemical calculations to study the thermodynamic driving force (the hydricity) and the kinetic barriers (the nucleophilicity) of HT and the efficiency of catalyst regeneration. Although the hydricity and nucleophilicity of 1,2-PyH₂ have been calculated in previous studies,^{14,19,24} we report here improved results obtained through a more rigorous approach. We also examine the effects of functionalization on these (dihydro)pyridinic derivatives, listed in Figure 1, with electron-withdrawing groups (EWGs) and electron-donating groups (EDGs). On the basis of this compilation of calculated thermodynamic and kinetic data, we have identified candidate molecules that are predicted to readily catalyze CO_2 reduction in water.

In addition to calculating the thermodynamic and kinetic strengths of DHP catalysts, we computationally determine their efficiency as recyclable catalysts by calculating reduction potentials and pK_a values of reaction intermediate species and by predicting reaction pathways through which these catalysts are formed. The feasibility of routes to form the active hydride donors is discussed, as is the ability of these donors to effect chemical reductions, which we determined by calculating the thermodynamics and kinetics of the various elementary steps, including the HT to hydride acceptors of interest, in particular

CO_2 (Scheme 1); indeed, we will focus on calculating the ability of various pyridinic hydrides to transfer a hydride to

Scheme 1. Possible Routes for the Electrochemical Formation of Hydride Donors XH_2 by the Reduction of X and the Reduction of a Target Oxidant Y, Such as CO_2 , HCOOH , and H_2CO , via Hydride and Proton Transfers to and from XH_2 ^a



^aProton and electron/hydride transfers are shown in blue and red, respectively. The route in black indicates sequential PT–ET–PT–ET reduction of X to XH_2 , whereas the dashed green route represents possible proton-coupled electron transfer (PCET) steps. $\text{pK}_{\text{a},1}$ and $\text{pK}_{\text{a},2}$ correspond to the pK_a values of XH^+ and $\text{XH}_2^{+\bullet}$, respectively. Similarly, E_1^0 and E_2^0 represent the reduction potentials of XH^+ and $\text{XH}_2^{+\bullet}$.

CO_2 in aqueous solvent.¹⁹ Given our focus on HT to CO_2 in water, our analysis of hydrides requires their aqueous hydricities, in contrast to the more commonly reported acetonitrile hydricity, as a descriptor of DHP thermodynamic strength; this is because solvation by water promotes stronger activity of DHP catalysts than does acetonitrile: due to the production of ionic products in HT reactions, DHPs are stronger hydride donors in the more polar, and thus more

Table 1. Summary of Hydricity in Acetonitrile $\Delta G_{\text{XH}_2, \text{a}}^0$, Aqueous Hydricity $\Delta G_{\text{XH}_2, \text{w}}^0$, Nucleophilicity N , Reduction Potential E^0 , and $\text{p}K_{\text{a}}$ Values Related to the Formation of and Hydride Transfer by DHPs^a

XH_2^b	R^c	Hydricity (a) ^d	Hydricity (w) ^e	N^f	E^0_{g}	E^0_{f} (PCET) ^h	E^0_{2}	E^0_{2} (PCET) ⁱ	$\text{p}K_{\text{a},1}^k$	$\text{p}K_{\text{a},2}^l$
1	H	50.5	19.9	10.5	-1.32	-1.15	0.29	0.28	5.14	1.05
	CONH ₂	59.8	28.1	9.2	-0.71	-0.78	-0.15	-0.56	1.89	-6.40
	CN	63.4	29.5	7.0	-0.56	-0.90	0.76	0.07	-1.08	-11.63
	CH ₃	47.8	17.0	11.1	-1.44	-1.22	0.23	0.28	6.23	2.11
	NH ₂	41.8	7.7	12.4	-1.73	-1.45	-0.17	0.15	6.79	6.13
2	H ^m	48.6	16.7	11.8	-1.32	-1.15	0.06	0.15	5.14	2.65
	CONH ₂	54.6	24.0	10.1	-0.79	-0.75	0.33	-0.08	2.79	-6.69
	CN	60.8	28.2	8.5	-0.49	-0.62	0.51	-0.16	0.57	-9.89
	CH ₃	45.8	15.2	12.3	-1.49	-1.24	-0.01	0.20	5.69	3.20
	NH ₂	37.4	3.1	13.9	-2.03	-1.54	-0.50	0.13	8.30	10.02
3	H	59.3	28.3	8.1	-0.97	-0.78	0.41	0.29	5.33	-0.75
	CONH ₂	64.8	35.7	6.6	-0.56	-0.44	0.54	0.07	1.87	-8.24
	CN	71.5	38.1	4.4	-0.27	-0.57	0.84	0.12	-1.16	-10.42
	CH ₃	56.6	26.9	8.9	-1.09	-0.82	0.34	0.28	5.53	-0.02
	NH ₂	48.3	13.1	10.8	-1.46	-1.12	-0.10	0.14	8.17	6.19
4	H	56.8	25.8	9.5	-0.97	-0.78	0.39	0.19	5.43	-3.31
	CONH ₂	60.9	30.0	9.0	-0.63	-0.56	0.55	0.05	3.88	-8.57
	CN	67.0	34.2	6.9	-0.26	-0.35	0.70	-0.10	1.69	-13.53
	CH ₃	54.2	24.5	10.1	-1.09	-0.83	0.33	0.18	5.54	-1.83
	NH ₂	44.9	10.2	11.6	-1.55	-1.03	-0.12	-0.01	9.29	-0.39
5	H	62.8	34.7	7.0	-0.74	-0.53	0.42	0.22	3.69	-3.35
	CONH ₂	67.7	41.1	5.5	-0.54	-0.38	0.53	0.18	0.93	-10.09
	CN	74.5	45.4	3.3	-0.12	-0.38	0.78	0.09	-3.78	-14.44
	CH ₃	59.3	31.1	7.8	-0.87	-0.56	0.35	0.19	5.27	-3.04
	NH ₂	50.2	24.5	10.1	-1.26	-0.87	0.10	0.21	5.32	2.14
6	H	61.3	31.7	8.2	-0.74	-0.53	0.46	0.15	4.80	-5.24
	CONH ₂	65.0	35.9	7.4	-0.47	-0.37	0.55	0.06	2.85	-8.30
	CN	70.5	39.7	5.8	-0.12	-0.19	0.65	-0.07	0.94	-13.53
	CH ₃	58.5	29.9	8.9	-0.85	-0.56	0.41	0.14	5.25	-4.81
	NH ₂	48.7	21.1	10.6	-1.24	-0.70	0.32	0.13	9.24	-4.97
7	H	69.3	39.5	4.9	-0.63	-0.34	0.51	0.36	5.73	-1.51
	CONH ₂	76.5	47.5	2.6	-0.37	-0.23	0.89	0.46	3.52	-6.07
	CN	83.2	49.7	0.6	-0.10	-0.14	1.07	0.42	2.63	-9.48
	CH ₃	66.4	37.3	5.9	-0.73	-0.40	0.36	0.38	6.76	0.80
	NH ₂	61.4	33.1	6.8	-0.89	-0.52	-0.24	0.33	6.98	8.83
8	H	64.0	31.4	7.5	-0.97	-0.81	0.47	0.42	4.87	-1.24
	CONH ₂	70.7	36.9	5.5	-0.60	-0.58	0.76	0.35	3.47	-6.81
	CN	76.9	42.7	3.8	-0.32	-0.47	1.01	0.36	1.20	-10.02
	CH ₃	61.2	27.9	8.5	-1.11	-0.87	0.25	0.36	5.30	1.49
	NH ₂	51.4	23.2	10.3	-1.52	-1.05	-0.05	0.53	8.53	9.60

^aHT reaction: $\text{XH}_2 + \text{CO}_2 \rightarrow \text{XH}^+ + \text{HCOO}^-$. ^bNumbers designate the family of DHP derivatives found in Figure 1: 1,4-dihydropyridine (1), 1,2-dihydropyridine (2), 1,4-dihydroquinoline (3), 1,2-dihydroquinoline (4), 1,4-dihydrobenzoquinoline (5), 1,2-dihydrobenzoquinoline (6), dihydroacridine (7), and dihydrophenanthridine (8). ^cFunctional groups attached to corresponding family of DHPs. ^dHydricity values (kcal/mol) calculated in acetonitrile solvent using the isodesmic approach (IA). ^eHydricity values (kcal/mol) calculated in water solvent using the $\text{p}K_{\text{a}}$ -isodesmic approach (PIA). Candidate molecules for CO_2 reduction in water are highlighted in blue, with $\Delta G_{\text{XH}_2}^w < 24$ kcal/mol. ^fNucleophilicity values calculated in dichloromethane (DCM) solvent. All reported $\text{p}K_{\text{a}}$ and E^0 values were computed in aqueous solvent at 298.15 K and 1 atm. ^gFirst reduction potential values (versus SCE) for the electron transfer $E^0(\text{XH}^+/\text{XH}^0)$. ^hFirst reduction potential values (versus SCE) for the net proton and electron transfer $E^0(\text{X}/\text{XH}^0)$. ⁱSecond reduction potential values (versus SCE) for the electron transfer $E^0(\text{XH}_2^+/\text{XH}_2)$. ^jSecond reduction potential values (versus SCE) for the net proton and electron transfer $E^0(\text{XH}^0/\text{XH}_2)$. ^k $\text{p}K_{\text{a}}$ values of XH^+ species. ^l $\text{p}K_{\text{a}}$ values of XH_2^+ species. ^mThe improved hydricity and nucleophilicity values of 2-H (1,2-PyH₂) should replace the numbers previously reported in ref 19.

stabilizing, aqueous media than in acetonitrile. In addition, solvation by water allows concerted, solvent-assisted HT–PT, and more favorable thermodynamics^{14,19,25} can supply hydrogen atoms through water splitting techniques^{26,27} and finally is environmentally friendly. Our analysis reveals a significant trade-off between catalyst strength and efficiency, and we suggest various applications of pyridinic derivatives that could possibly circumvent this restriction.

2. METHODS AND COMPUTATIONAL DETAILS

2.1. Computational Details. Quantum chemical calculations were used to evaluate hydricities, nucleophilicities, and reduction potentials of the molecules of interest. Density functional theory (DFT) based on the M06 exchange–correlation functional²⁸ combined with the 6-31+G(d,p) basis set was employed.²⁹ This level of theory should provide a reliable description of the properties of the molecules and reactions of interest, hydride transfers, because the M06 functional has been parameterized with experimental data for similar molecular systems.²⁸ To validate the M06/6-31+G(d,p) method, we performed calculations on a subset of five of the smallest hydrides (entries 1a–1e in Table 1) with the wB97XD/6-311++G(d,p) method.³⁰ All calculations were conducted within the Gaussian 16 software package.³¹

Solvation effects were incorporated using a conductor-like polarizable continuous model (CPCM)^{32,33} for water, the primary solvent of interest as explained in the Introduction, but also for acetonitrile and dichloromethane (DCM) solvents (as will be seen, different solvents are relevant for different properties); explicit inclusion of several hydrating water molecules is used for certain purposes. We obtained hydricities agreeing to within 3 kcal/mol for the two methods, with M06 tending to estimate lower hydricities, by approximately 2.5 kcal/mol relative to the wB97XD functional (see Table S4). Our methods also accurately reproduce experimental nucleophilicity, reduction potential, and pK_a values, as reported in the respective subsections which follow.

For all properties, and free energies in particular, we define the reference state of all reactions as the isolated reactants in solution.³⁴ The calculated vibrational force constants were used to compute the zero-point energy and vibrational entropy contributions to free energies and the vibrational heat capacity contributions to enthalpies at 298 K. Such force constants also confirmed that reactants and products have only positive vibrational modes and that transition states have only one imaginary mode, corresponding to hydride transfer to CO₂.

2.2. Methods. **2.2.1. Hydricity.** The thermodynamic hydricity of a dihydride XH₂ ($\Delta G_{\text{XH}_2, \text{a}}^0$), which is typically measured in acetonitrile solvent (hence the “a” subscript), quantifies a hydride donor’s strength via its heterolytic dissociation free energy of XH, i.e., the hydricity $\Delta G_{\text{XH}_2, \text{a}}^0$ of XH₂ is defined as the heterolytic dissociation free energy, so that more negative values indicate greater hydride donor strength^{35–37}



$$\Delta G_{\text{XH}_2, \text{a}}^0 = G_{\text{XH}^+} - G_{\text{XH}_2} + G_{\text{H}^-} \quad (4)$$

Current DFT-based methods do not generally calculate the free energy of the solvated hydride G_{H^-} accurately.³⁸ Therefore, we have instead employed two different indirect approaches to estimate the hydricity: (1) the isodesmic

approach (IA)^{39,40} and (2) the linear scaling approach (LSA) proposed by Muckerman.⁴¹ The former avoids a free-energy calculation for the solvated hydride ion, but instead requires knowledge of the experimentally determined hydricity value of a reference molecule. We select 1-benzyl-1,4-dihydronicotinamide (BNAH) as reference because it has a structure very similar to the dihydropyridinic molecules that we consider. The difference between eq 4 and its equivalent for the hydride dissociation of BNAH yields the isodesmic eq 5, in which the G_{H^-} term is eliminated. Equation 5 enables the IA estimation of the hydricity of XH₂ via calculation of the free energies of BNAH, BNA⁺, XH₂, and XH⁺.

$$\begin{aligned} \Delta G_{\text{XH}_2, \text{a}}^0 &= \Delta G_{\text{BNAH}, \text{a}}^0 + (G_{\text{XH}^+} - G_{\text{XH}_2}) \\ &\quad + (G_{\text{BNAH}} - G_{\text{BNA}^+}) \end{aligned} \quad (5)$$

For a test set of eight hydrides, we find that the IA predicts hydricities more accurately than does the LSA (details are provided in Supporting Information (SI) Section B). The mean absolute deviations between experimental and calculated hydricities are approximately 0.5 and 2 kcal/mol for the IA and LSA, respectively. Accordingly, we will employ IA hydricity values for the thermodynamic screening of DHPs.

2.2.2. Aqueous Hydricity. Hydricity values are most commonly reported in acetonitrile solvent and thus provide only a limited description of hydride donor strength. As noted in the Introduction, we are more interested in reactions of DHPs in water solvent and as a result require the aqueous hydricity ($\Delta G_{\text{XH}_2, \text{w}}^0$) to measure the thermodynamic strength of these hydrides. Unfortunately, the determination of the aqueous hydricity is more challenging due to the lack of experimental data for benchmarking calculations. To validate the use of the IA, we propose an alternative approach to the calculation of $\Delta G_{\text{XH}_2, \text{w}}^0$ which contains elements from the IA as well as the well-known experimental potential-pK_a approach and show in Table S3 that both methods are equally reliable in producing aqueous hydricities for DHPs.³⁸ We begin with the reaction of the dihydride XH₂ with CO₂ in water to produce X and formic acid



with the reaction free energy

$$\Delta G_{\text{rxn}, \text{w}}^0 = G_{\text{X}} + G_{\text{HCOOH}} - G_{\text{XH}_2} - G_{\text{CO}_2} \quad (7)$$

This equation can be manipulated, using the definitions of pK_a(KH⁺), pK_a(HCOOH), $\Delta G_{\text{XH}_2}^0$, and $\Delta G_{\text{HCOO}^-}^0$ to relate the free energy of reaction to the hydricities of XH₂ and HCOO[−] via eq 8 and its equivalent, eq 9 (see also SI Section S.B3)

$$\begin{aligned} \Delta G_{\text{rxn}} &= (G_{\text{X}} + G_{\text{H}^+} - G_{\text{XH}^+}) + (G_{\text{HCOOH}} - G_{\text{HCOO}^-} \\ &\quad - G_{\text{H}^+}) + (G_{\text{XH}^+} + G_{\text{H}^-} - G_{\text{XH}_2}) \\ &\quad + (G_{\text{HCOO}^-} - G_{\text{CO}_2} - G_{\text{H}^-}) \end{aligned} \quad (8)$$

$$\begin{aligned} \Delta G_{\text{rxn}, \text{w}}^0 &= 1.36\text{pK}_{\text{a}}(\text{XH}^+) - 1.36\text{pK}_{\text{a}}(\text{HCOOH}) \\ &\quad + \Delta G_{\text{XH}_2, \text{w}}^0 - \Delta G_{\text{HCOO}^-, \text{w}}^0 \end{aligned} \quad (9)$$

We have selected the reaction that transfers a hydride and proton from the dihydride to CO₂, eq 6, for the approach that employs eq 9, which we will refer to as the “pK_a-isodesmic” approach (PIA) because formic acid and the formate

intermediate have pK_a and hydricity values, which are well defined in the literature. Equation 9 requires the pK_a value of formic acid (3.8)⁴² as well as formate's aqueous hydricity (24 kcal/mol).⁴³ As we will discuss, thermodynamically capable catalysts possess $\Delta G_{XH_2,w}^0$ (w = water) lower than that of formate in water solvent; formate has a hydricity of 43 kcal/mol⁴⁴ in acetonitrile and 24 kcal/mol in water.^{43,45} Depending on the availability of reliable pK_a data, the PIA could outperform the IA in terms of accuracy. Otherwise, the IA is a more straightforward method for obtaining aqueous hydricities from a computational chemist's standpoint.

2.2.3. Nucleophilicity. The nucleophilicity N describes the strength of hydride donors in terms of the rates of their HT reactions to hydride acceptors, and is a kinetic property, unlike the hydricity, which is a thermodynamic property. Mayr and co-workers proposed that N is directly related to the log of the rate constant of the nucleophile–electrophile HT reaction via eq 10,^{46–49}

$$\log k_{20^\circ\text{C}} = s_N(N + E) \quad (10)$$

where s_N is the nucleophile-dependent sensitivity parameter and E is the electrophilicity parameter for the hydride acceptor. We use the methodology proposed in our previous work to estimate N , based on a linear fit based on the relation between N and the activation free energy of a HT reaction shown in eq 11²⁴

$$N = \frac{1}{2.3s_NRT} \Delta G^\ddagger + N_0 \quad (11)$$

This relation assumes that s_N and E are constant for the HT reactions of interest. We ensure the constancy of E by selecting CO_2 as the electrophile for all of the reactions considered. Furthermore, s_N is approximately constant for all reactions considered here because the hydride donors all belong to a single family, carbon-based aromatic heterocyclic hydrides. All reaction activation free energies are calculated in DCM solvent because the nucleophilicity scale is more established for DCM than for aqueous solvents. Consequently, N values will be used as an indicator of the kinetic activity of DHPs relative to 1,2-PyH₂, which has been extensively studied in aqueous media in earlier works.¹⁹ Because these DHP catalysts presumably have a similar HT mechanism with CO_2 as the electrophile, we can compare their relative strengths in DCM and assume similar behavior in water solvent; support for this claim is given in SI Section S.C5. We take $\Delta G^\ddagger = 20$ kcal/mol as the upper limit, for which reactions can proceed at reasonable rates at room temperature;^{14,21} as shown in SI Section S.C4, this corresponds to $N = \sim 9$ for HT from DHPs to CO_2 . Additional details for the methodology, including the treatment of reaction activation entropy, are provided in SI Section C.

2.2.4. Calculation of pK_a Values. The aqueous hydricity calculation via eq 9 requires aqueous solution pK_a values. We calculate these, after Schlegel,⁵⁰ from the deprotonation reaction $\text{AH} \rightleftharpoons \text{A}^- + \text{H}^+$, using eqs 12 and 13

$$pK_a = \frac{\Delta G_{\text{aq}}}{2.303RT} \quad (12)$$

$$\Delta G_{\text{aq}} = G_{\text{A}^-} + G_{\text{H}^+} - G_{\text{AH}} \quad (13)$$

This approach requires the free energy of a proton in water G_{H^+} , determined experimentally to be -265.9 kcal/mol.^{51–54}

The remaining species' free energies required in eq 13 for pK_a calculations are computed using a hybrid explicit/implicit solvation model. Five explicit water molecules are configured around the relevant solute and that cluster is embedded in CPCM- H_2O implicit solvent; this approach directly describes important hydrogen bonding interactions as well as the far-field solvation effects to provide a reasonable description of hydration effects in HT thermodynamics.⁵⁰ Table S9 demonstrates that this approach reproduces experimental pK_a values more accurately than does the CPCM-only model, with a mean absolute deviation from experiment of 0.33 pK_a units. The hybrid solvation model was applied only for pK_a calculations because it reproduced experimental data accurately, whereas benchmarking tests indicated inaccurate reaction free energies (for $\Delta G_{XH_2,w}^0$ calculations) and reduction potentials (SI Tables S5 and S8).⁵⁵

2.2.5. Reduction Potential. The efficiency of DHP catalysts can be quantified by the reduction potential required to generate these catalysts from their inactive pyridinic form. Efficient catalysts are characterized by less negative reduction potentials, thus requiring minimal free-energy expenditure for catalyst regeneration. The calculation of reduction potentials E^0 (vs SCE) in water follows the approach proposed by Tossell, which uses eqs 14 and 15⁵⁶

$$\Delta G_{\text{red}} = G_{\text{XH}} - G_{\text{XH}^+} \quad (14)$$

$$E^0 = \frac{-100.5 - \Delta G_{\text{red}}}{23.05} - 0.24 \quad (15)$$

We obtain good agreement of our calculated reduction potentials with reported experimental values: the mean average deviation = 0.14 V, see SI Table S7. Reduction potential calculations for proton-coupled electron transfer are explained in detail in SI Section S.D2.

3. RESULTS AND DISCUSSION

3.1. DHP Formation. We first evaluate the effects of functionalizing pyridinic derivatives with electron-donating and electron-withdrawing groups on catalyst effectiveness, specifically on pK_a and reduction potential values, parameters related to catalyst recyclability. In particular, the pK_a values provide clues of probable pathways (e.g., protonation) by which dihydropyridines are formed from their respective pyridinic forms, whereas the reduction potential values indicate the efficiency of these hydrides as catalysts. Scheme 1 indicates that electrochemical reduction of pyridinic derivatives X to their active dihydropyridinic (DHP) forms XH_2 can be performed through sequential PT–ET–PT–ET reactions or via proton-coupled electron transfers (PCETs) depending on the pK_a values of protonated intermediates. For our present purposes, we define PCET as the concerted transfer of separate proton and electron species; other definitions are possible.⁵⁷

High pK_a values of XH^+ and $\text{XH}_2^{+\bullet}$ favor proton retention and thus allow for sequential PT–ET reduction, whereas very low pK_a values indicate that due to their acidity, these species are thermodynamically disfavored or even inaccessible in practice. For such cases, the PCET route (indicated by green dashes in Scheme 1) is preferred over sequential PT–ET because PCETs avoid the formation of thermodynamically disfavored species XH^+ and $\text{XH}_2^{+\bullet}$ by concertedly transferring an electron and a proton to X or XH^0 . Moreover, $pK_{a,2}$ values for $\text{XH}_2^{+\bullet}$ indicate which variants of DHPs are more easily

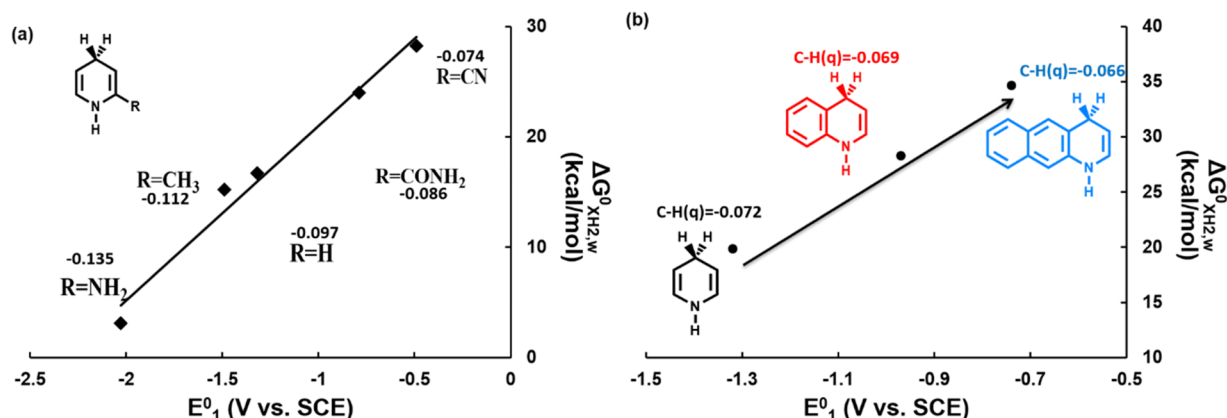


Figure 2. Aqueous hydricity (kcal/mol) vs the first reduction potential E_1^0 (V versus SCE). (a) Effects of functionalizing 1,2-dihydropyridine (entry 2-H, Table 1) with EDGs (R = NH₂ and CH₃; lower $\Delta G_{\text{XH}_2}^0$ and more negative E_1^0) and EWGs (R = CN and CONH₂; larger $\Delta G_{\text{XH}_2}^0$ and less negative E_1^0). (b) Comparison of multiple-ring systems: 1,4-dihydropyridine (one ring; entry 1-H, Table 1) with 1,4-dihydroquinoline (two rings; entry 3-H, Table 1) and 1,4-dihydrobenzoquinoline (three rings; entry 5-H, Table 1). The arrow indicates the direction of increasing number of fused aromatic rings, which have an effect similar to that of the EWGs in (a). Lower magnitude hydricities (y-axis) represent stronger hydrides, whereas more negative reduction potentials (x-axis) indicate less energy-efficient catalysts. The charge density of the hydridic hydrogen calculated via atomic polar tensor (APT) population analysis⁵⁹ is displayed in each DHP derivative.

formed: larger values indicate more protonated species, which are available for reduction, as in the cases of 1,4-PyH₂ and 1,2-PyH₂ (1-H versus 2-H, Table 1).

Here, we consider the effects of functionalizing dihydropyridines with electron-donating and electron-withdrawing groups on the pK_a and E^0 values to determine the overall influence these groups would have on XH₂ catalyst regeneration. A promising catalyst would optimally (i) possess protonated intermediates XH⁺ and XH₂^{•+} with sufficiently high pK_a values (relative to the pH of the solution) to facilitate formation of XH₂ via PTs in the PT–ET–PT–ET route, (ii) reduction potentials that are not too negative, thus not requiring excessive free energy to access the active hydride state XH₂, and (iii) possess $N > 9$ to be an adequately strong hydride donor to perform HT reductions at practical rates.

The first reduction of XH⁺ to XH⁰ requires significantly more energy (that is a more negative reduction potential E_1^0) for the hydrides considered here than does the second reduction E_2^0 of XH₂^{•+} to XH₂. Given that this might appear surprising because a cation is reduced in both reductions, we discuss this in some detail. The first reduction dearomatizes XH⁺ and creates a radical species. The high free energy required to dearomatize XH⁺ in turn supplies the driving force for HT, driven by rearomatization of the dihydropyridinic species XH₂ in forming XH⁺.¹⁹ In contrast, the second reduction to form XH₂ involves no dearomatization because XH₂^{•+} is not aromatic and instead only reduces a radical cation to a closed shell species. Consequently, once the applied potential is sufficiently negative to initiate the first reduction, the second reduction following protonation of XH⁰ proceeds with relative ease.¹⁴ A similar trend is observed experimentally by Cao et al. for dihydropyridine-type compounds.⁵⁸

The calculated E^0 and pK_a values associated to sequential ETs and PTs to form hydrides from their aromatic state X are compiled in Table 1 and exhibit a clear trend as we now discuss. Functionalizing DHPs with electron-donating groups (EDGs) produces species with more negative reduction potentials, as shown in Figure 2a, larger pK_a values and as also seen in Figure 2a ultimately lower hydricities (stronger hydrides). In contrast, functionalization with electron-with-

drawing groups (EWGs) leads to species with lower pK_a values and less negative reduction potentials, the latter as also seen in Figure 2a. Concerning the pK_a effects, EDGs increase the electron density in the conjugated π -system, allowing for easier protonation of X and XH⁰ species, which explains the larger pK_a values relative to EWG-functionalized species. The effects on E_1^0 are related to the increased electron density contributed by EDGs, which also destabilizes the reduced π -system of XH. As a result, the addition of the 7th electron into the aromatic ring that disrupts the aromaticity must also overcome repulsive Coulombic interactions with the EDG-provided electron density. Thus, reduction of XH⁺ in the case of EDG-functionalized DHPs requires more energy (produces lower reduction potentials) than does reduction of XH⁺ for EWG-functionalized DHPs. In addition to these inductive effects on reduction potentials, the size of the π -space affects the reduction potentials of XH⁺; larger π -systems containing multiple fused aromatic rings, such as quinoline (3-H, Table 1) and benzoquinoline (5-H, Table 1), delocalize the electrons in the π -space, thus causing effects similar to EWGs to produce less negative reduction potentials, as shown in Figure 2b, that also result in lower pK_a values. These trends can be intuitively derived based on the chemistry of EDGs and EWGs and therefore validate the methods used to obtain pK_a and E^0 values.

3.2. DHP Strength. We have calculated, via the methods discussed in Section 2.2, the hydricities (in acetonitrile), aqueous hydricities, and nucleophilicities of DHPs, summarized in Table 1, to analyze the impact of functionalization on hydride donor strength. The driving force for HT reactions by DHPs is rearomatization; upon hydride donation, DHPs regain their aromaticity. The thermodynamic and kinetic strengths of DHPs are governed by the electron densities on the active ring or, more precisely, on the hydridic hydrogen; EDGs increase the electron density in the ring, creating stronger hydrides with lower $\Delta G_{\text{XH}_2}^0$ and larger N , whereas EWGs have the opposite effect. These interpretations are supported by the charges on the hydride calculated via atomic polar tensor (APT) population analysis⁵⁹ shown in Figure 2 (see also: Figure S10). Similar to pK_a and E^0 , the trends in thermodynamic and

kinetic properties of hydride donors functionalized by EDGs and EWGs were expected and thus validate the methods employed to derive the hydricity and nucleophilicity parameters.

The HT reaction characterized by $\text{XH}_2 + \text{CO}_2 \leftrightarrow \text{XH}^+ + \text{HCO}_2^-$ favors the products with larger $\Delta G_{\text{XH}_2, \text{w}}^0$. Consequently, the forward reaction to generate formate by HT to CO_2 is favored in water if $\Delta G_{\text{XH}_2, \text{w}}^0$ of XH_2 is lower than that of formate ($\Delta G_{\text{XH}_2, \text{w}}^0(\text{HCO}_2^-) = 24 \text{ kcal/mol}$ for the reaction of $\text{HCO}_2^- \rightarrow \text{CO}_2 + \text{H}^-$).⁴³ Out of 40 DHPs, 10 are considered in this work and are identified as thermodynamically competent in reducing CO_2 , possessing hydricities $\Delta G_{\text{XH}_2, \text{w}}^0 < 24 \text{ kcal/mol}$, and are highlighted “blue” in Table 1. In contrast, only 2 DHPs from the studied set are deemed capable of CO_2 reduction in acetonitrile, with $\Delta G_{\text{XH}_2, \text{a}}^0 < 43 \text{ kcal/mol}$.

We also find, as shown in Figure 3, that the DHPs' thermodynamic and kinetic parameters correlate linearly. Such

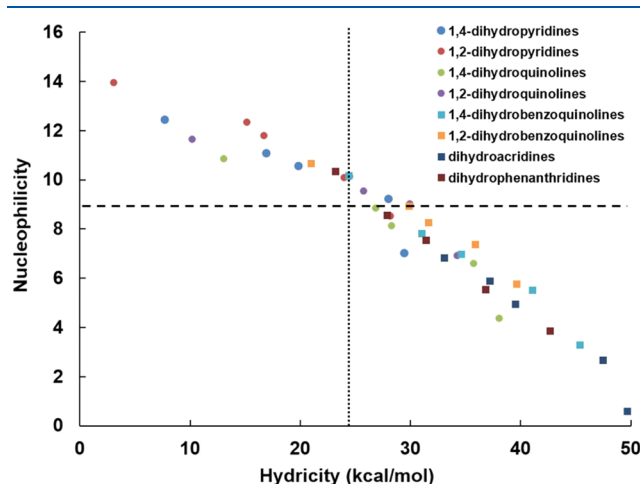


Figure 3. Predicted nucleophilicity N values plotted against predicted aqueous hydricity values (kcal/mol). The plot shows a clear linear free-energy relation for HT reactions within the DHP class of hydrides. The dotted vertical line represents the aqueous hydricity of formate (24 kcal/mol). The dashed horizontal line crosses the y -axis at $N = 9$, an approximate threshold indicating the kinetic competence of a catalyst at 298 K. DHPs in the upper-left quadrant are considered capable of reducing CO_2 ($N > 9$ and $\Delta G_{\text{XH}_2, \text{w}}^0 < 24 \text{ kcal/mol}$).

trends within the same class of molecules are often identified in physical organic chemistry as quantitative structure–property (or activity) relationships or more broadly linear free-energy relationships.^{60–64} In Section 2.2, we indicated that $N = 9$ is the minimum N value required for practical kinetics of CO_2 reduction by DHPs. Figure 3 shows that all thermodynamically capable DHPs ($\Delta G_{\text{XH}_2, \text{w}}^0 < 24 \text{ kcal/mol}$) are also kinetically strong ($N > 9$). As such, the thermodynamic criterion is sufficient in determining a DHP catalyst's capability in reducing CO_2 .

An analogous linear relationship, $\Delta G^\ddagger = \alpha \Delta H + \beta$, described by the Evans–Polanyi principle between the activation free energy and enthalpy change of the hydride-transfer reaction in two different solvents, water and acetonitrile, is demonstrated in Figures S7 and S8.⁶¹ It is worth noting that even though these solvents produce considerably different thermodynamics, the change in the α parameter of the Evans–Polanyi relationship between the two solvents is less than 2%. This

indicates that the hydride-transfer barriers drop by approximately the same amount in changing the solvent from acetonitrile to water.

3.3. $\Delta G_{\text{XH}_2, \text{w}}^0 - E_1^0$ Correlation. Although our analysis above identified many hydrides that possess hydricities and nucleophilicities that indicate their capability to reduce CO_2 , the recycling of the catalyst remains a considerable challenge. In this subsection, we consider this issue and suggest strategies, which may circumvent this limitation.

Our results collected in Figure 4 demonstrate that a strong correlation exists between the first (and most free-energy

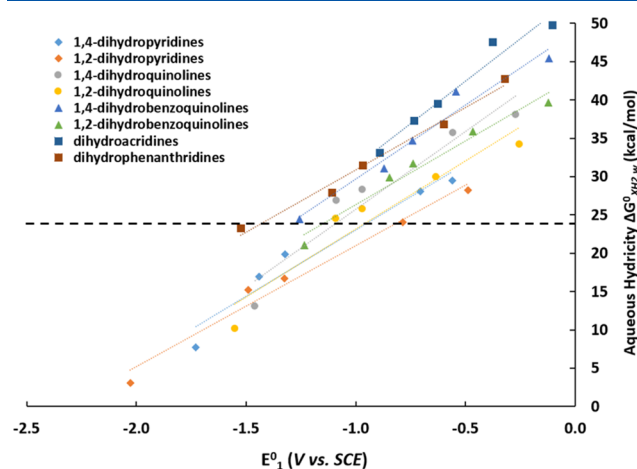


Figure 4. Aqueous hydricity $\Delta G_{\text{XH}_2, \text{w}}^0$ (kcal/mol), defined by the reaction shown in eq 6 in water solvent versus the first reduction potential E_1^0 (V versus SCE), defined by eqs 14 and 15, for all 40 DHPs considered in this work. Linear regression is performed to obtain the dotted lines for each family of DHPs. The dashed horizontal line corresponds to formate's aqueous hydricity $\Delta G_{\text{XH}_2, \text{w}}^0 = 24 \text{ kcal/mol}$. As discussed in the text, a trade-off (linear within each DHP family) exists between the strength of the hydride (increasing with smaller $\Delta G_{\text{XH}_2, \text{w}}^0$) and the free energy required to recycle it (increasing with more negative E_1^0). The best catalysts would optimally lie in the lower region of figure as much as possible toward the right.

demanding) reduction potential E_1^0 and the aqueous hydricities $\Delta G_{\text{XH}_2, \text{w}}^0$ of DHPs. We have also indicated in Figure 4 that only all DHPs with $\Delta G_{\text{XH}_2, \text{w}}^0 < 24 \text{ kcal/mol}$ (the hydricity of formate) are theoretically capable of reducing CO_2 . This $\Delta G_{\text{XH}_2, \text{w}}^0 - E_1^0$ correlation, which is approximately linear for each DHP type, reveals the essential recycling challenge: stronger catalysts require more free energy to recycle them to produce their active hydride forms. In fact, this type of correlation is not limited to pyridinic systems, but applies to the majority of hydride donor species.³⁸

Overcoming the restrictions resulting from the $\Delta G_{\text{XH}_2, \text{w}}^0 - E_1^0$ correlation would be of significant help in the effort to achieve conversion of CO_2 into fuels effected by molecular catalysts. We will suggest various possible approaches below that might avoid such a $\Delta G_{\text{XH}_2, \text{w}}^0 - E_1^0$ correlation. Although heterogeneous CO_2 catalysis possibly involving surface-bound pyridines is similarly also of great interest,⁶⁵ here we limit ourselves in offering general suggestions, some based on recent work,^{66,67} for improving homogeneous CO_2 reduction efficiency.

As discussed in Section 3.2, the strength of the hydride donor, which stems from its rearomatization,¹⁹ scales linearly with the free energy required to dearomatize the XH^+ species. Because the $\Delta G_{\text{XH}_2, \text{w}}^0 - E_1^0$ correlation in pyridines is fundamentally governed by aromaticity effects ($\Delta G_{\text{XH}_2, \text{w}}^0$ and E_1^0 correlate to rearomatization and dearomatization, respectively), one might be tempted to avoid the correlation's difficulty by completely avoiding aromatic compounds in the search for molecular CO_2 reduction catalysts. However, a more nuanced solution might lie in suitably changing the nature of the hydride rather than simply evading aromaticity. The search for catalysts need not be limited to carbon-based hydrides, but could also include other classes of potentially regenerable hydrides, such as nitrogen-based hydrides (i.e., nitrogen-bound hydride). An encouraging example is provided by the biological cofactor flavin adenine dinucleotide (FAD), which commonly participates as a hydride-transfer reagent in its fully reduced FADH_2 state;⁶⁸ remarkably the nitrogen-based FADH_2 's hydricity is on par with that of the carbon-based cofactor NADPH.³⁹ Moreover, FAD reduction requires a less negative reduction potential than does NADPH, further motivating consideration of nitrogen- or flavin-based hydrides.^{38,69} In this connection, our group recently predicted that the carbon-based hydride of flavin derivative 6,7-dimethyl-4-hydroxy-2-mercaptopteridine is thermodynamically capable of reducing CO_2 , but is kinetically slow.²¹ However, the electron density arguments discussed in Section 3.2 indicate that functionalizing analogs of pteridines with EDGs would likely favorably affect the HT kinetics by increasing the hydride's electron density.

A more pronounced difference in approach for overcoming the restrictions imposed by the $\Delta G_{\text{XH}_2, \text{w}}^0 - E_1^0$ relationship would be the use of transition-metal molecular catalysts, as illustrated by the ruthenium-based molecular catalyst examined by Tanaka and co-workers.^{70–76} These authors showed that the Ru complex with the organic ligand 1,5-dihydro-2-(2-pyridyl)-benzo[*b*]-1,5-naphthyridine (which is analogous to pyridine) catalyzes HT to CO_2 for formate production via this catalysis with the assistance of carboxylate bases. Additionally, Tanaka's catalyst was found to be recycled purely photochemically.⁷³ Muckerman and co-workers proposed that the Tanaka catalyst is sufficiently strong in reducing CO_2 after accessing higher reduced states via photoexcitation.^{41,77,78} Photochemistry's role in the Tanaka system is then to produce a highly reducing electron in an excited state of the molecular catalyst by photoexcitation to aid the CO_2 reduction. Although ruthenium is expensive, and thus likely not economic for industrial-scale CO_2 reduction applications, understanding the Tanaka catalyst's nature can provide key insights into novel alternative strategies for reducing CO_2 and catalyst regeneration.

Another approach to CO_2 reduction could involve devising systems, where photoelectrochemical phenomena are exploited without involving prohibitively expensive metals. For example, a photocatalyst might act to regenerate DHPs by shuttling electrons from an electrode to the pyridinic forms *X*. Here, long-lived high-energy excited states of the photocatalyst possessing highly reducing electrons could be accessed by photoexcitation, thus overcoming the difficulty that direct electrochemical photocatalyst reduction occurs at considerably more negative potentials. In conjunction with this approach, a molecular bridge between pyridine and the photocatalyst could

be utilized to facilitate electron transfer to pyridine, an approach often used for charge transfer in nanoparticle systems.^{79,80}

Finally, in addition to improving the hydride donor strength, CO_2 activation needs to be investigated. Various molecular approaches have been shown to be effective in activating CO_2 for reduction; here, we mention a few selected examples to discuss CO_2 activation. Rhodium pincer catalysts can activate CO_2 via a Rh–O bond, allowing easier HT to the carbon atom to yield formate. This catalyst, however, lacks the ability to transfer protons and allow further reduction to more desirable products, such as methanol.⁶⁶ Frustrated Lewis pairs (FLPs) have also been shown activated CO_2 to enhance the thermodynamics and kinetics of HT to CO_2 from ammonia-borane, although the irreversible binding of FLPs to CO_2 and the reduced products hinders the production of more useful products.⁶⁷ FLP chemistry, involving the simultaneous activation of CO_2 and the reducing agent, is a promising approach, as shown by Fontaine and co-workers.⁸¹ Most CO_2 reductions by FLPs, however, require HTs from hydroborane or hydrosilane reagents, and some recyclability issues will need to be overcome.

4. CONCLUDING REMARKS

Quantum chemical methods based on DFT have been exploited herein to quantify the effects of solvation and functionalization on the hydride donor strength of dihydropyridine species and the feasibility of their formation. We have focused on hydride transfer (HT) to CO_2 , of importance for both energy and environmental concerns. We also focus on HT reactions in water because HT by dihydropyridines is thermodynamically and kinetically more favorable in water than in acetonitrile, for example, due to solvent stabilization of the ionic-reduced products, specifically hydrogen bonding between the solvent and products. The hydricity and nucleophilicity concepts are used to evaluate hydrides' thermodynamic and kinetic activities, which stem from the stabilizing effects of rearomatization. The reduction potential associated to dearomatizing pyridines (and recycling the active form of the catalyst) is used as metric for catalyst efficiency. To better understand dihydropyridine catalysts, a systematic study was performed to determine the effect of functional groups on the hydricity, nucleophilicity, and reduction potentials of pyridinic derivatives.

An important aspect of our findings is that electron-donating groups create stronger hydrides, as indicated by the lower hydricity values and higher electron densities localized on the hydridic hydrogen. This advantage, however, comes at the expense of larger free-energy requirements (more negative reduction potentials) to regenerate the dihydropyridinic catalysts; this effect is due to the feature that the pyridinic ring, now with a larger electron density in the π -system contributed by the electron-donating groups, must be dearomatized. Multiple-ring systems and electron-withdrawing functional groups, in contrast to electron-donating groups, produce more efficient, but weaker hydrides, as the electron density is delocalized across the molecule. This HT strength–efficiency relationship between the catalyst hydricity and the free energy required to recycle it restricts the capability of such recyclable organic hydrides for reductions by HT and thus presents a major challenge for the design of hydrides to catalyze CO_2 reduction. We have presented several suggestions

for circumventing this limitation, which could prove to be of interest in the catalytic conversion of CO₂ into fuels.

■ ASSOCIATED CONTENT

■ Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jpcc.8b08536.

Computational methods; transition-state corrections; detailed derivation of eq 9; benchmarking calculations of pK_a, reduction potential, nucleophilicity, and hydricity; charge density analysis; molecular coordinates and zero-point energies for all hydrides (PDF)

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Notes

The authors declare no competing financial interest.

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